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BCA (Sem-III) – Operating

Roll - 71

System (302) Core-VI

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

1. Choose the correct alternative for each of the following questions : $2 \times 10 = 20$

(a) What is the ready state of a process ?

→(i) When process is scheduled to run after some execution

(ii) When process is unable to run until source task has been completed

(iii) When process is using the CPU

(iv) None of the above

(b) What is short term scheduler ?

(i) It selects which process has to be brought into the ready queue

→(ii) It selects which process has to be executed next and allocates CPU

(iii) It selects which process has to be removed from memory swapping

(iv) None of the above

(c) Round Robin Scheduling falls under the category of :

- (i) Non-preemptive
 - (ii) Preemptive
 - (iii) Both (i) & (ii)
 - (iv) None of the above
- (d) A minimum of ____ variable is/are required to be shared between processes to solve the critical section problem :
- (i) One
 - (ii) Two
 - (iii) Three
 - (iv) Four
- (e) In unix each process is identified by its :
- (i) PCB
 - (ii) Device queue
 - (iii) Process identifies
 - (iv) None of the above

(f) Device Driver required in ?

(i) Register

(ii) Main memory

(iii) Disk

(iv) Cache

(g) Thread is not shared among which of the following ?

(i) Stack

(ii) Program counter

(iii) Both program counter and stack

(iv) None of these

(h) Main memory of a computer system is ?

(i) Non-volatile

(ii) Volatile

(iii) Restricted

(iv) Unrestricted

(i) OS classifies the threads as :

(i) Mainframe and motherboard level

(ii) Kernel and user level

(iii) Security and memory level

(iv) OS and CPU level

(j) FIFO scheduling is a type of :

(i) Pre-emptive scheduling

(ii) Non-pre-emptive scheduling

(iii) Deadline scheduling

(iv) None of the above

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. Define time sharing operating system.

3. What do you mean by context switching ?
4. What are the various layers of file system ?
5. Write short notes on fire wall.
6. Write short notes on processor and user modes.
7. Define thread issues.
8. Define memory allocation.
9. What is paging ? Explain.

Section-C

Long Answer type Questions

Answer any two questions of the following :

15×2=30

10. What is operating system ? What is the need for an operating system ? Discuss the major function of an OS.

11. What are the various threats to security of a system ? Elaborate in details the different mechanism of protection and security.

✓ 12. Explain process scheduling ? Write non-pre-emptive and pre-emptive scheduling algorithm.

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